

Sushil Khyalia

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EDUCATION

Carnegie Mellon University

Master of Science in Machine Learning, GPA: 4.12/4.00

Pittsburgh, PA

December 2024

Indian Institute of Technology Bombay

Bachelor of Technology in Computer Science with Honors, GPA: 9.00/10.00

Mumbai, India

June 2020

WORK EXPERIENCE

Sarvam AI | ML Researcher

LLM Team

Bangalore, India

April 2025 - Present

- Core contributor on the 15-person team that trained Sarvam 30B and 105B, India's first open-source reasoning MoE models ([blog](#)), now powering Sarvam's Samvaad and Indus products
- Built the asynchronous GRPO stack decoupling generation, reward computation, and policy updates, bounding trajectory staleness against policy age for stability at high GPU utilization
- Moved policy optimization to a group-relative, CISPO-inspired objective with no KL anchoring to a reference policy, training to convergence without reward collapse
- Designed verifiable RL environments over math, coding, STEM, web search, and tool use, single- and multi-turn, with execution-based rewards alongside rubric graders for instruction adherence and format
- Built the RL curriculum, filtering prompts by pass rate against early checkpoints and allocating rollouts by information gain as a knapsack problem under a fixed generation budget
- RL post-training drove the 105B's agentic and reasoning results, topping its comparison set on Tau2 (68.3) and BrowseComp (49.5), with 71.7 LiveCodeBench v6 and 88.3 AIME25 rising to 96.7 with tools
- Curated 10B+ synthetic agentic-data tokens for the late-stage pre-training mixture
- Built the SFT and RLVR pipelines for Sarvam-M, a 24B hybrid-reasoning model ([blog](#)), spanning GRPO curriculum, hardness-proxy prompt sampling, and reward engineering (+21.6% math, +17.6% code)

Carnegie Mellon University | Research Assistant

Advisor: Prof. László A. Jeni

Pittsburgh, PA

May 2023 - August 2023

- Designed an alignment layer projecting video and audio features into an LLM's text-token space for end-to-end multimodal fine-tuning
- Curated MAVERIX (AAAI 2026), a 2,556-question audio-visual benchmark requiring joint cross-modal inference

Samsung Research | Engineer

Language and Voice Team, Data Analysis Lab

Seoul, South Korea

October 2020 - March 2023

- Derived closed-form forward and backward moments for every transformer component, giving a unified signal-propagation theory of exploding gradients, rank collapse, and attention-induced instability (ICML 2024)
- Proposed DeepScaleLM, an initialization and residual-scaling scheme conserving unit moments through depth, enabling stable training to 1000 layers and deeper-narrower models that beat larger shallow baselines at half the parameters
- Designed a lexicon-free G2P scheme using HuBERT k-means clusters as phoneme units (ICASSP 2024), and contrastive objectives for transferable graph representations (DLG-AAAI 2022)

SELECTED PUBLICATIONS

Transformers Get Stable: An End-to-End Signal Propagation Theory for Language Models

Forty-first International Conference on Machine Learning (ICML 2024)

[link](#)

PAC Mode Estimation using PPR Martingale Confidence Sequences

25th International Conference on Artificial Intelligence and Statistics (AISTATS 2022)

[link](#)

Meta-Learning for Effective Multi-task and Multilingual Modelling

16th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2021)

[link](#)

MAVERIX: Multimodal Audio-Visual Evaluation and Recognition Index

Fortieth Annual AAAI Conference on Artificial Intelligence (AAAI 2026)

[link](#)

SELECTED RESEARCH PROJECTS

Classifier-guided Unlearning in Diffusion Models: Steered diffusion models away from target concepts using concept-classifier feedback

Structure of the Policy Space of 2-action MDPs: Characterized the policy space via Acyclic Unique Sink Orientations, proving Howard's Policy Iteration optimal among deterministic algorithms for 3- and 4-state instances